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Experiment 10: 5G Network Slicing Environment Using MATLAB

Aim : To analyze the performance of a 5G Network Slicing environment by studying the number of network slices, slice utilization, and end-to-end latency using MATLAB.

Objectives :

Objective – 1: Analyze the Number of Network Slices Analyze the creation and allocation of multiple network slices in a 5G network using MATLAB. Observe how different slices can be configured to support diverse service requirements and applications.

Objective – 2: Compare Slice Utilization (%) Compare the utilization of different network slices using MATLAB. Analyze how efficiently network resources are allocated and utilized across multiple slices.

Objective – 3: Analyze End-to-End Latency (ms) Analyze the end-to-end latency of different network slices using MATLAB. Compare the latency values to understand the Quality of Service (QoS) requirements of various 5G applications.

Procedure:

Objective 1

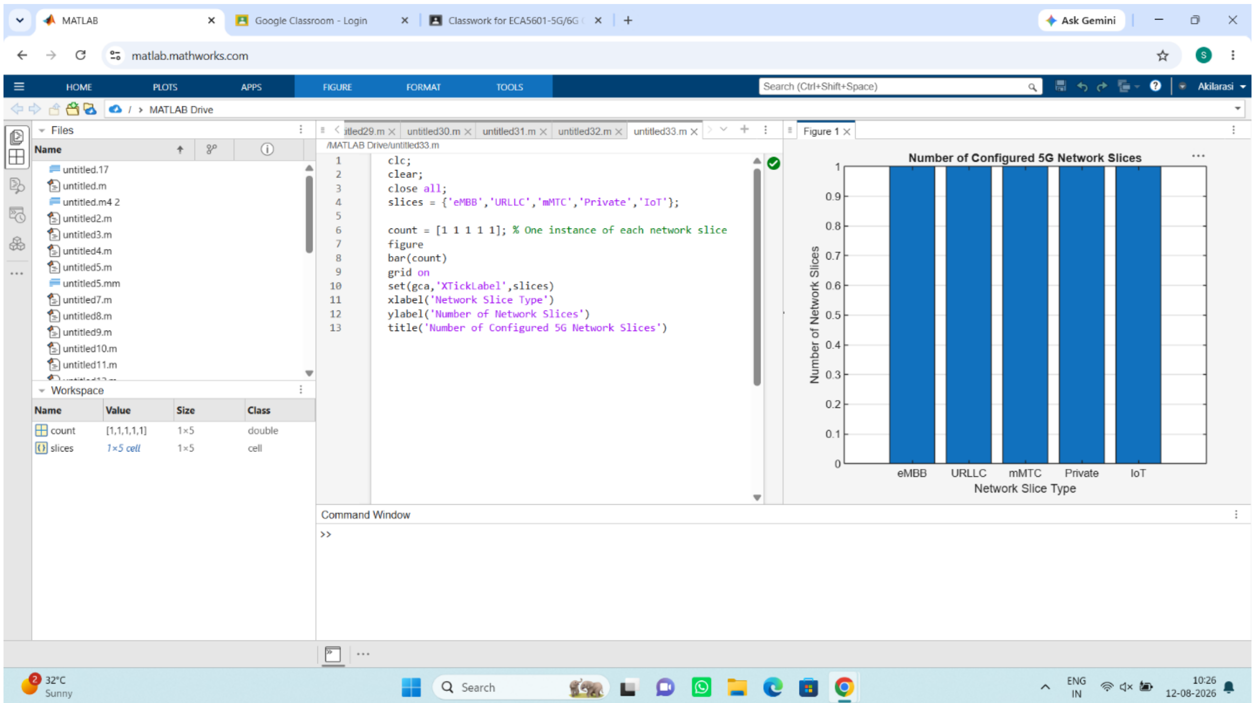
```
clc;

clear;

close all;

slices = {'eMBB','URLLC','mMTC','Private','IoT'};

count = [1 1 1 1 1];

figure

bar(count)

% One instance of each network slice

grid on

set(gca,'XTickLabel',slices)
```

xlabel('Network Slice Type')

ylabel('Number of Network Slices')

title('Number of Configured 5G Network Slices')

Output:



Objective 2:

clc;

clear;

close all;

slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});

utilization = [85 70 60 75 50];

figure;

bar(slices, utilization);

grid on;

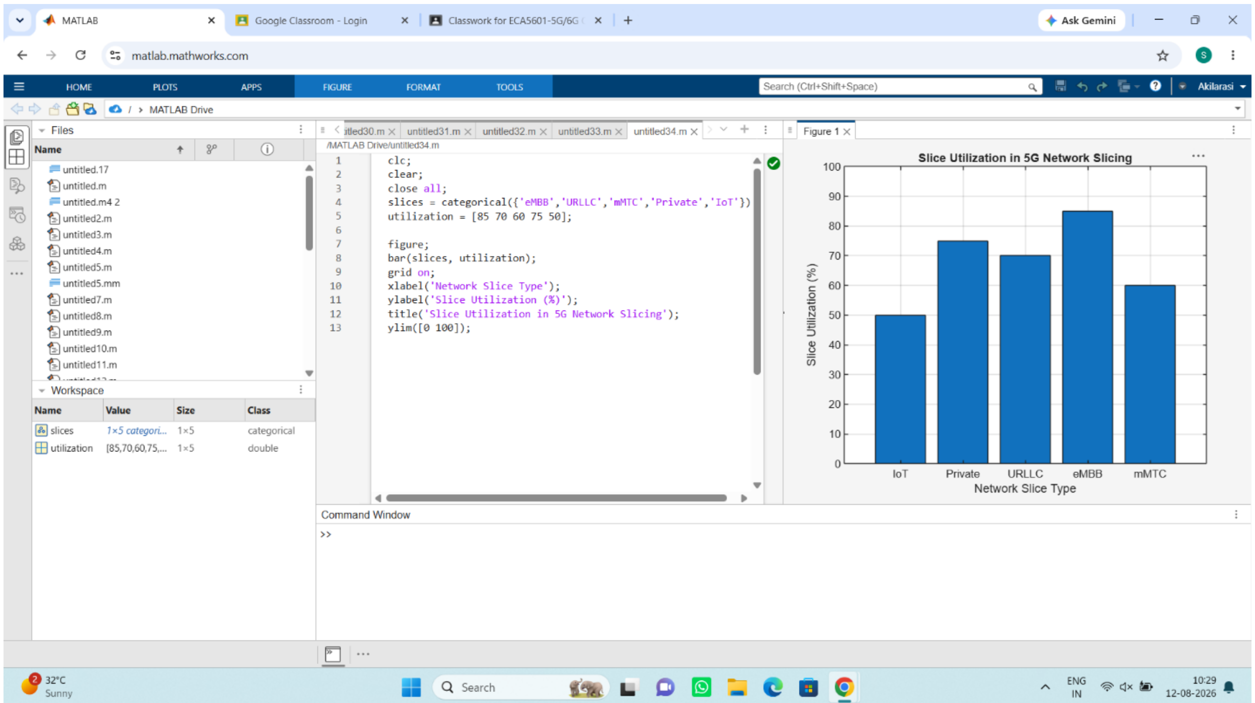
xlabel('Network Slice Type');

ylabel('Slice Utilization (%)');

title('Slice Utilization in 5G Network Slicing');

ylim([0 100]);

Output:



Objective 3:

clc;

clear;

close all;

slices = [1 2 3 4 5];

latency = [25 5 40 15 60];

figure;

plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);

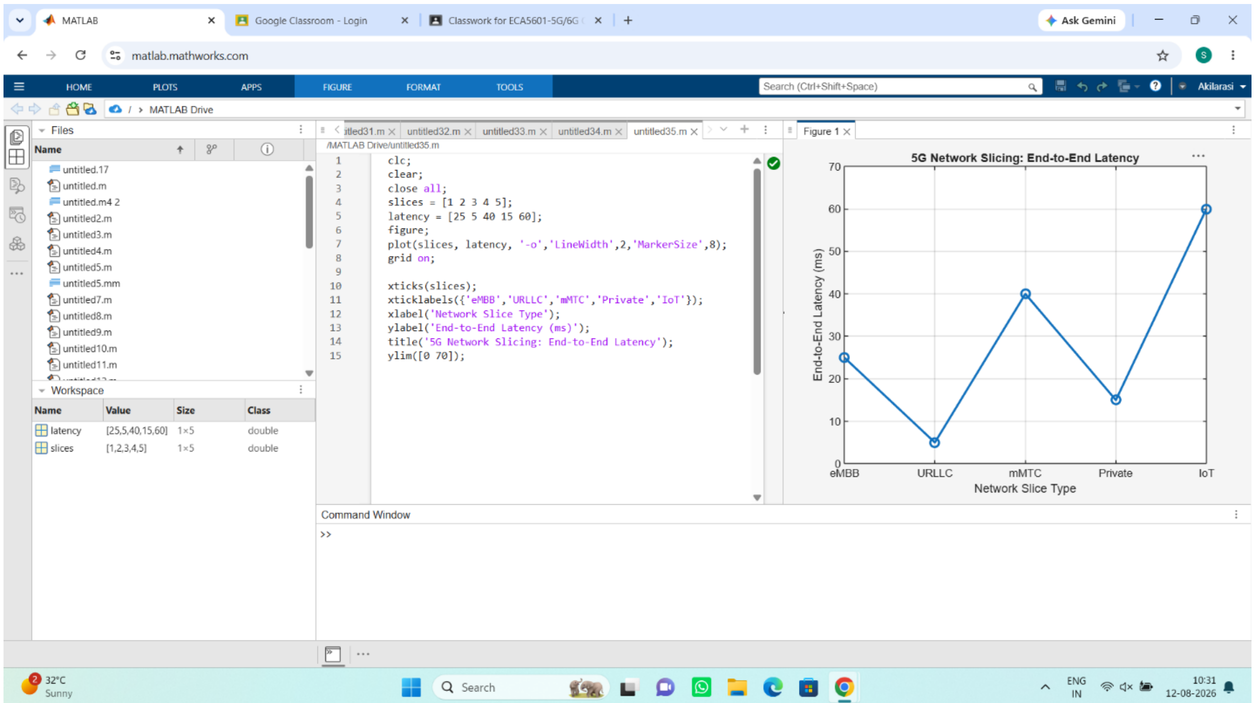
grid on;

xticks(slices);

xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});

```
xlabel('Network Slice Type');

ylabel('End-to-End Latency (ms)');

title('5G Network Slicing: End-to-End Latency');

ylim([0 70]);
```

Output:



Result: The MATLAB analysis successfully demonstrated the characteristics of a 5G Network Slicing environment. The results showed that multiple network slices can coexist while exhibiting different utilization levels and latency characteristics based on their intended services.